Exercise Solutions

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Exercise 1.

A topological space is said to be Hausdorff if for any two distinct points in the space, there exists two disjoint open sets containing each point. Every metric space is automatically a Hausdorff space (see if you can prove this for yourself), but plenty non-Hausdorff spaces exist.

(Difficulty 2/10)

(a) Show that if a finite group G acts on a Hausdorff space Y such that no nontrivial elements has a fixed point, the action is even. Recall that this means that for all $y \in Y$, there exists an open set U containing y such that the open sets $\{gU\}_{g \in G}$ are *pairwise disjoint*, i.e. $g_1 U \cap g_2 U = \varnothing$ for all $g_1, g_2 \in G$. By the result in lecture, this implies that if Y is connected then $Y \to G \setminus Y$ is a Galois cover.

If you are not comfortable with working with topological spaces, you may assume that Y is a metric space instead.

Proof.
Pick some $y \in Y$. As the space is Hausdorff, for every $gy \neq y$ there exists disjoint open sets V_e, V_g of y and gy respectively. Then by iterating over G we obtain the finite set $\left\{V_g\right\}_{g \in G}$. As the finite intersection of open sets is open, the set $U := \bigcap_{g \in G} g^{-1} V_g$ is an open set which contains the point y; as each open set $g^{-1} U_g$ contains the point $g^{-1} gy = y$.

I claim that the set $\{gU\}_{g \in G}$ is pairwise disjoint as desired. Indeed, if $g_0 \in G$, then

$$g_0 U = \bigcap_{g \in G} g_0 g^{-1} U = \bigcap_{g \in G} (g g_0^{-1})^{-1} U = \bigcap_{g g_0^{-1} \in G} (g g_0^{-1})^{-1} U = U.$$

It remains to show that distinct elements of G produce distinct open sets, i.e. if $gU = g'U$ then $g = g'$. To that end it is enough to show that if $gy = g'y$ then $g = g'$. But if $gy = g'y$, we would have that $gg'^{-1} y = y$. But no nontrivial element can have a fixed point, so $gg'^{-1} = e$ and $g = g'$. ∎

(Difficulty 4/10)

(b) Show that the finiteness of G is a necessary requirement, by considering the example where $Y = S^1 = \{(\cos(t), \sin(t)) \in \mathbb{R}^2 \mid t \in [0, 2\pi)\}$ is the unit circle and $G = \mathbb{Z}$; where $n \in \mathbb{Z}$ acts on Y via rotation by n radians counterclockwise; i.e. $n \cdot (\sin(t), \cos(t)) = (\sin(n + t), \cos(n + t))$. You need not verify that this is a well-defined group action, and you can take for granted the group also acts by *isometries*; i.e. $d(p, q) = d(n \cdot p, n \cdot q)$ for all $p, q \in S^1$ and $n \in \mathbb{Z}$.

Proof.

Fix the point $p = (\cos(0), \sin(0))$. Then we get the infinite sequence $\{p, 1 \cdot p = (\cos(1), \sin(1)), 2 \cdot p = (\cos(2), \sin(2)), ...\}$. Every point in this sequence must be distinct, as $\cos(n) = \cos(m)$ if and only if $n - m = 2k\pi$. But then S^1 is a compact set by Heine-Borel, so the sequence must have some convergent subsequence $\{n_1 \cdot p, n_2 \cdot p, n_3 \cdot p, ...\}$ with limit point q. Hence for all $\varepsilon > 0$ there exists n_i, n_j such that $d(n_i \cdot p - q) < \varepsilon/2$ and $d(n_j \cdot p - q) < \varepsilon/2$. By the triangle inequality

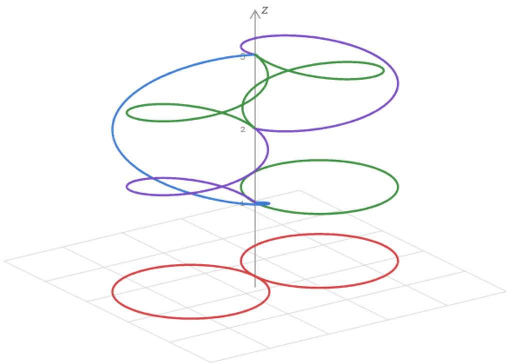
$$d(n_i \cdot p, n_i \cdot p) \leq d(n_i \cdot p - q) + d(n_j \cdot p - q) = \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Moreover, as $\mathbb{Z}$ is acting by isometries, we have that $d(p, (n_j - n_i) \cdot p) = d(n_i \cdot p, n_i \cdot p)$. But then the action cannot be even, as any open set U containing p_0 has to contain some point of the form $(n_j - n_i) \cdot p$; such that it cannot be disjoint from the set $(n_j - n_i)U$. ■

Exercise 2.

(Difficulty 4/10)

Show that the following cover of the *figure 8* (denoted $S_1 \vee S_1$) is *not* Galois:



Proof.
A *cut point* of a topological space is a point $x \in X$ such that the subspace $X \setminus \{x\}$ is disconnected. Notice that in the fiber above the crossing point $c \in S^1 \vee S^1$, the bottom crossing (the point $(0, 0, 1)$) is a cut point, while none of the other two points above c are cut points. Hence it is sufficient to prove the following lemma:

If X is a topological space with automorphism ϕ, then if $x \in X$ is a cut point of X then so is $\phi(x)$.

Write $X \setminus \{x\} = U \sqcup V$ as a union of disjoint nonempty open (or closed) sets. Then as automorphisms are bijections, $\phi(X \setminus \{x\}) = X \setminus \{\phi(x)\} = \phi(U) \sqcup \phi(V)$. As automorphisms are bicontinuous maps, $\phi(U)$ and $\phi(V)$ are both nonempty open (closed) maps as well. Hence $\phi(x)$ is a cut point of X as well. ■

Exercise 3.

(Difficulty 6/10)

Show that a connected cover $Y \to X$ is Galois with automorphism group G if and only if the fibre product $Y \times_X Y$ is isomorphic to the trivial cover $Y \times G \to Y$; where G is assumed to carry the discrete topology.

Proof.
$(\Rightarrow)$: Suppose that $Y \times_X Y$ is isomorphic to $Y \times G$ as Y-covers via some ϕ. Clearly we have that G acts simply transitively on $Y \times G$ by multiplication on the right coordinate, so we can define an induced action of G on $Y \times_X Y$ via pre-composing with ϕ. As ϕ is a homeomorphism the induced action is also necessarily simply transitive.

$(\Leftarrow)$: Suppose that Y is Galois. Then G acts transitively on the fibers, such that $\pi_X^{-1}(y)$ is equal to the set $\{(y, gy)\}_{g \in G}$. Hence we get a well-defined bijection $[\phi : Y \times_X Y \to Y \times G$.

We check that ϕ is continuous: Let $U \times V$ be a basic open of $Y \times G$. Then $\phi^{-1}(U \times V) = \{(y, gy) : y \in U, g \in V\} = \bigcup_{g \in G} U \times_X gU$, which is seen to be open.

We check that ϕ is open: Let $U \times_X U'$ be a basic open set of $Y \times_X Y$. Then $\phi(U \times_X U')$ is seen to be a basic open set in $Y \times G$ as well. Since G carries the discrete topology we only need verify the image be open on the Y-coordinate. But then that is exactly U. ■

Exercise 4.

(Difficulty 10/10)

Let $S^1 = \{(\cos(t), \sin(t)) \in \mathbb{R}^2 \mid t \in [0, 2\pi)\}$ denote the unit circle.

Show that every cover $p : S^1 \to S^1$ is Galois, and find its Galois group.

(If you are not comfortable with working with topological spaces, you may assume that every cover is already a metric space.)

(**Hint.** There are many ways to approach this problem. I outline one of them below, but you are welcome to try and figure out your own method.

(1) Suppose that $p : S^1 \to S^1$ is some degree n cover. Construct a parameterization $f : [0, 2n\pi) \to S^1$ such that $(p \circ f)(t) = (\cos(t), \sin(t))$.

(2) Show that $f(t) \mapsto f(t + 2\pi)$ induces a covering automorphism of S^1 over S^1.

(3) Show that there are exactly n distinct automorphisms, and how this implies that the cover is Galois. What is the group structure?)

Proof.
Suppose that $p : S^1 \to S^1$ is a degree n cover. Parameterize the upper circle via $[0, 2n\pi)$, and parameterize the lower circle via $[0, 2\pi)$.

(1) Define a function $g : [0, 2n\pi) \to [0, 2\pi)$ as follows: Start with $u \in [0, 2n\pi)$, which corresponds to the point $x = (\cos(u), \sin(u))$. Apply the projection p to obtain some other point $p(x) \in S^1$ in the lower circle, which will correspond to some $v \in [0, 2\pi)$ under the parameterization. We define $g(u) := v$.

It follows by construction that g is a piecewise continuous strictly increasing function with jump discontinuities at the points where $g(u) = 2k\pi$. By shifting the pieces of g if necessary, we obtain a strictly increasing continuous function $g : [0, 2n\pi) \to [0, 2n\pi)$. As the inverse of a strictly increasing continuous function is continuous, via composing with g^{-1} we can reparameterize to get $f : [0, 2n\pi) \to S^1$ as desired.

(2) That adding 2π defines a homeomorphism of $[0, 2n\pi)$ with $[2\pi, 2n\pi + 2\pi)$ is clear. Under the parameterization this gives us an automorphism of S^1. So we only need to verify that this is a cover automorphism. But then $(p \circ f)(t + 2\pi) = (\cos(t + 2\pi), \sin(t + 2\pi)) = (\cos(t), \sin(t)) = (p \circ f)(t)$.

(3) As $t \mapsto t + 2\pi$ is a cover automorphism, so is $t \mapsto t + 2k\pi$ for all $0 \leq k < n$, so this gives n distinct cover automorphisms. But then as distinct automorphisms must act distinctly on any point (theorem from lecture), this implies that the action is transitive: Fix a point y below, and fix a point $x \in p^{-1}(y)$. Then the n different automorphisms above must send x to n different elements in the fiber $p^{-1}(y)$, but there are at most n choices to send x to. So the cover is Galois.

Finally, as the size of the automorphism group is at most the degree of the cover (theorem from lecture), every automorphism is of the form $t \mapsto t + 2k\pi$. We see that the Galois group is $\mathbb{Z}/n\mathbb{Z}$, where the isomorphism is $(t \mapsto t + 2k\pi) \mapsto k + n\mathbb{Z}$. ∎

Alternate proofs of the question are on the next page.

Proof (continued).

Here is a proof that uses complex analysis:

Identify the image of p with the unit circle inside the complex numbers $\mathbb{C}$. Then p can be thought of as a contour integral that starts and ends at 1. Then by the residue theorem, $\frac{1}{2\pi i}\oint_p \frac{1}{z}dz = n$ for some integer n (by reversing p if necessary WLOG assume that $n > 0$). By Cauchy's (homotopy) integral theorem, this implies that we can reparameterize the contour integral as $t \mapsto e^{2\pi int}$. The rest of the proof follows as above from (2) and (3). ■

Here is a proof that uses algebraic topology:

The universal cover of S^1 is $\mathbb{R}$. By the universal property of the universal cover, the map $\pi : \mathbb{R} \to S^1$ factors (uniquely) through the cover $S^1 \to S^1$. One can easily verify that the universal cover is Galois with Galois group isomorphic to $\mathbb{Z}$. As $\mathbb{Z}$ is abelian, every subgroup is normal; so under the correspondence this implies that *every* cover of S^1 is Galois. Moreover, under the correspondence, degree n covers correspond to degree n quotients of $\mathbb{Z}$. But then this is classified uniquely to be $\mathbb{Z}/n\mathbb{Z}$. ■

If you have only listened to my first talk, do *not* attempt the questions after this page (you're welcome to take a look - they will just be nigh impossible to do without the theorems I will introduce in the second talk)**.**

The second half of this document is meant to be done only after the second talk.

Warning: I have tried my best to keep the questions at a manageable difficulty level for those with a minimal mathematical background. Despite my best efforts, the rest of this document represents a *significant* difficulty jump in the exercises. You are regardless welcome to attempt them and ask me questions if you need help.

Exercise 5.

(Difficulty 14/10)

(a) Let X be a connected and locally path-connected, but not necessarily locally simply connected topological space. Show that every finite cover of X is *dominated* by some Galois cover; i.e. for every finite cover $\pi : Y \to X$ there exists a cover $Z \to Y \to X$ such that $Z \to X$ is Galois.

(**Hint.** Prove the question statement via the following steps:

(1) First argue why you can WLOG assume that Y is connected.

(2) Consider the n-fold fibered product of Y over X, denoted Y^n_X. Notice that this space admits a natural action of S_n.

(3) Let $\hat{Y}$ denote the connected component of Y^n_X that contains the point $(y_1, ..., y_n)$, where $\{y_i\} = \pi^{-1}(x)$ is the fiber above some chosen basepoint x. This space still admits the same action of S_n.

(4) Notice that $\pi_1(X, x)$ embeds as a transitive subgroup of S_n via the monodromy action, and S_n has a well-defined morphism into $\mathrm{Aut}\big(\hat{Y}|X\big)$. Argue why this implies that $\hat{Y}$ is Galois.)

Proof.
Fix some base point $x \in X$, and WLOG suppose that Y is a connected cover of degree n such that $p^{-1}(x) = y_1, ..., y_n$. Now notice that the n-fold product Y^n_X is a degree n^n cover of X, such that it admits a natural action of S_n of automorphisms via permutation of the coordinates (it is a well-defined deck transformation as coordinate permutations are automorphisms of $Y^n \supseteq Y^n_X$).

Let $\hat{Y}$ denote the connected component of the n-fold product Y^n_X that contains the point $(y_1, ..., y_n) \in Y \times_X ... \times_X Y$. Note that as the coordinates are distinct, this space admits the same action of S_n. As $\hat{Y}$ is connected by assumption, the monodromy action of $\pi_1(X, x)$ on $\hat{Y}$ induces a morphism $\phi : \pi_1(X, x) \to S_n \to \mathrm{Aut}\big(\hat{Y}|X\big)$ such that the image $H := \phi(\pi_1(X, x))$ is a transitive subgroup of S_n. But then the degree of $\hat{Y}$ is exactly equal to the size of the $\pi_1(X, x)$ orbit on the fiber (as connected, locally path connected spaces are necessarily path connected), so ϕ is in fact surjective on $\mathrm{Aut}\big(\hat{Y}|X\big)$. We conclude that $\hat{Y}$ is Galois. To finish, notice that $\hat{Y}$ is a finite cover of Y via projection onto the first coordinate. ∎

(Difficulty 14/10)

(b) Show that the Galois cover you have constructed is *terminal* among all the Galois covers of Y over X: Namely, if $W \to Y$ is any Galois cover, then there exists a unique morphism $W \to Z$ such that the following diagram commutes:

$$\begin{tikzcd} W \arrow[d, dashed] \arrow[dr] & & \\ Z \arrow[r] & Y \arrow[r, "\pi"] & X \end{tikzcd}$$

(this is an instance of a *universal property*.)

Hence, we see that every finite cover of a space admits a *Galois closure*, which corresponds to the similar notion that we have for finite separable field extensions.

Proof.

Suppose that Z is some Galois cover of X such that its projection factors factors through a cover morphism $f : Z \to Y$. Fix some z_0 such that $f(z_0) = y_1$, and choose $g_i \in \mathrm{Aut}(Z|X)$ such that $f(g_i z_0) = y_i$ (take $g_1 = e$). Then we can define a morphism $\Psi : Z \to Y_X^n$ to be given by $z \mapsto (f(g_1 z), ..., f(g_n z))$ such that $\Psi(z_0) = (y_1, ..., y_n) \in \hat{Y}$. By the connectedness of Z we see that $\Psi(Z) \subseteq \hat{Y}$, and by Lemma 2.2.11. we see that Ψ is in fact a covering map. ■

Exercise 6.

(Difficulty 18/10)

A *topological group* is a group G that is also a topological space, such that the group structure is compatible with the topology in the following sense: 1) group multiplication is a continuous function $G \times G \to G$ (where $G \times G$ is viewed with the product topology), and 2) inversion is a continuous function $G \to G$.

Examples of topological groups include $\mathbb{R}^n$ under addition, or the circle group S^1 under multiplication.

Let G be a connected and locally simply connected topological group with unit element e. Let $\pi : \tilde{G}_e \to G$ be a universal cover, and $\tilde{e}$ the universal element in the fiber $\pi^{-1}(e)$.

(a) Equip $\tilde{G}_e$ with a natural structure of a topological group with unit element $\tilde{e}$ for which π becomes a homomorphism of topological groups.

Proof.
Recall that the universal cover $\tilde{G}_e$ is constructed as the set of paths starting at e up to (fixed end point) homotopy. Hence, we define a composition law on $\tilde{G}_e$ as follows: given $f, g : [0,1] \to G$, we let $fg(t) := f(t)g(t)$. That this is indeed a well-defined path follows from the continuity of the group operation and connectedness of G. That this defines a group law with identity element $\tilde{e}$, that π is a group homomorphism and that the diagram below commutes is a route verification. It remains to show continuity of the group operations.

$$\begin{tikzcd} \tilde{G}_e \times \tilde{G}_e \arrow[r, dashed, "\tilde{m}"] \arrow[d, "\pi \times \pi"'] & \tilde{G}_e \arrow[d, "\pi"'] \\ G \times G \arrow[r, "m"] & G \end{tikzcd} \qquad \begin{tikzcd} \tilde{U} \times \tilde{V} \arrow[r, mapsto, dashed] \arrow[d, mapsto] & \widetilde{W} \arrow[d, mapsto] \\ U \times V \arrow[r, mapsto] & W \end{tikzcd}$$

Suppose that $\tilde{f}, \tilde{g} \in \tilde{G}_e$ such that $\tilde{f}\tilde{g} = \tilde{h}$. Let $\widetilde{W}$ denote some open simply connected neighbourhood of $\tilde{h}$. It is then sufficient to show that $\left(\tilde{f}, \tilde{g}\right) \in \tilde{G}_e \times \tilde{G}_e$ admits a basic open neighbourhood contained in $\tilde{m}^{-1}\left(\widetilde{W}\right)$. To that end, by the continuity of multiplication in G, there exists $(f, g) \in U \times V \subseteq m^{-1}(W)$. By shrinking the neighbourhoods U, V if necessary we can WLOG assume they are simply connected and a local trivialization of $\pi \times \pi$. Hence by taking an appropriate lift there exists $\left(\tilde{f}, \tilde{g}\right) \in \tilde{U} \times \tilde{V}$.

It is then sufficient to show that $\tilde{m}\left(\tilde{U} \times \tilde{V}\right) \subseteq \widetilde{W}$. To that end, note that points in $\tilde{U} \times \tilde{V}$ are of the form $\left(\tilde{\alpha} \circ \tilde{f}, \tilde{\beta} \circ \tilde{g}\right)$ where $\tilde{\alpha}, \tilde{\beta}$ are paths contained in $\tilde{U}, \tilde{V}$ with startpoints at $\tilde{f}(1)$ and $\tilde{g}(1)$ respectively. Then $\tilde{m}\left(\tilde{\alpha} \circ \tilde{f}, \tilde{\beta} \circ \tilde{g}\right) = \tilde{\alpha}\tilde{\beta} \circ \tilde{f}\tilde{g}$. But then $\alpha\beta$ is a path contained in $U \times V$, such that its image is contained in W. By passing through the lift π, we see that the path $\tilde{\alpha}\tilde{\beta}$ must be contained in $\widetilde{W}$, and we are done.

The proof for showing that inversion is a continuous operation is near-identical, and we omit it here for the sake of brevity. ■

(Difficulty 16/10)

(b) Show that a group structure with the above properties is unique.

> ***Proof.***
> Suppose that m, m' are two group structures on $\tilde{G}$ that satisfy the given requirements. Then the following must be a commuting diagram of topological groups:
>
> $$\begin{tikzcd} \tilde{G} \times \tilde{G} \arrow[r, shift left, "m'"] \arrow[r, shift right, "m"'] \arrow[dr, dashed] & \tilde{G} \arrow[d, "\pi"] \\ & G \end{tikzcd}$$
>
> but then we are immediately done (theorem from lecture 1), as $\pi(m'(\tilde{e}, \tilde{e})) = \pi(m(\tilde{e}, \tilde{e})) = e$. ■

(Difficulty 16/10)

(c) Construct an exact sequence of topological groups

$$1 \to \pi_1(G, e)^{\mathrm{op}} \to \tilde{G}_e \to G \to 1,$$

the topology on $\pi_1(G, e)$ being discrete.

> ***Proof.***
> From (a) we have that π and the embedding of $\pi_1(G, e)^{\mathrm{op}}$ into $\tilde{G}_e$ are well-defined group homomorphisms. As a covering map it is necessarily surjective; and it is immediate that its kernel is $\pi_1(G, e)^{\mathrm{op}}$. We verify that the SES agrees with the given topologies: It is sufficient to show that the topology endowed on $\pi_1(G, e)^{\mathrm{op}}$ as a subspace of $\tilde{G}_e$ is discrete. But as then there is a natural identification of the points in $\pi_1(G, e)^{\mathrm{op}}$ with the elements of $\mathrm{Aut}\left(\tilde{G}_e \,|G\right)$ (theorem from lecture 2); and the action of $\mathrm{Aut}\left(\tilde{G}_e \,|G\right)$ on $\tilde{G}_e$ is even (theorem from lecture 1), we see that the induced subspace topology on $\pi_1(G, e)$ is exactly the discrete topology. ■